

Quartiles Definition

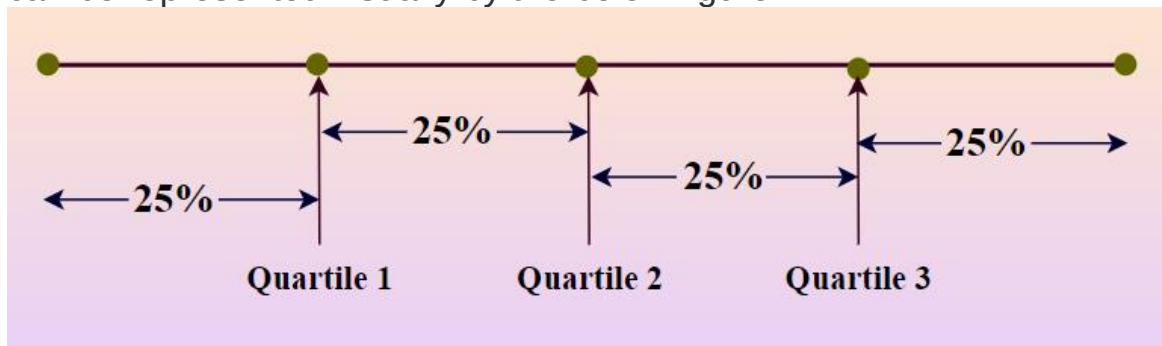
Quartiles are the values from the dataset which divide the dataset into four equal parts where each part of the dataset contains an equal number of observations.

There are three quartiles such as,

- First or Lower Quartile
- Second Quartile or [Median](#)
- Third or Upper Quartile

What is the Quartile Formula?

As mentioned above **Quartile divides the data into 4 equal parts**. This can be represented visually by the below figure.



- Quartile 1 lies between the starting term and the middle term.
- Quartile 2 lies between the starting terms and the last terms i.e., the Middle term.
- Quartile 3 lies between quartile 2 and the last term.

There is a separate formula for finding each quartile value and the steps to obtain the quartile formula are as shown below as follows:

- **Step 1:** Sort the given data in ascending order.
- **Step 2:** Find respective quartile values/terms as per need from the below formulae.
- **First Quartile** = $(\{n + 1\}/\{4\})^{th} \text{ term}$
- **Second Quartile** = $(\{n + 1\}/\{2\})^{th} \text{ term}$
- **Third Quartile** = $(\{3(n + 1)\}/\{4\})^{th} \text{ term}$

Where n is the total count of numbers in the given data.

Quartiles in Statistics

We know that the Median divides the data into two equal parts, in the same way, the **quartile divides the data into four parts**. Similar to the median which divides the data into half so that 50% of the estimation lies below the median and 50% lies above it, the quartile splits the data into i.e.,

- **First Part of Data:** From smallest to largest of numbers 25% of the value, comes under this part and also this part lies below the first quartile.
- **Second of Data:** Value between 25% and 50% of the data comes under this part and this part lies between the first and second quartile (Median).
- **Third of Data:** Value between 50% and 75% of the data comes under this part and this part lies between the second and third quartile.
- **Fourth of Data:** Greatest 25% of all values in the data comes under the fourth part and this part lies above the fourth quartile.

Interquartile Range

Interquartile Range is the **distance between the first quartile and the third quartile**. It is also known as a mid-spread. It helps us to calculate variation for the data which is divided into quartiles. The formula for calculating the Interquartile range is given by,

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1$$

Where,

- Q_3 is third/upper quartile, and
- Q_1 is first/lower quartile.

Quartile Deviation

Quartile Deviation is defined as half of the distance between the first quartile and the third quartile. It is also known as Semi Interquartile Range. The formula for quartile deviation is given by,

$$\text{Quartile Deviation} = (Q_3 - Q_1)/2$$

Find the first, second, and third Quartile for the data 8, 5, 15, 20, 18, 30, 40, 25

Solution:

Step 1: Sort the given data in the ascending order

5, 8, 15, 18, 20, 25, 30, 40.

Step 2: Find all Quartiles step by step

First Quartile = $\{(n + 1)/4\}$ th term

Here $n = 8$ because there are total 8 numbers in the given data.

$\Rightarrow$ First Quartile = $\{(8 + 1)/4\}$ th term

$\Rightarrow$ First Quartile = $\{9/4\}$ th term

$\Rightarrow$ First Quartile = 2.25th term

Thus, 2.25th Term = 2nd term + (0.25)(3rd term – 2nd term)

$\Rightarrow$ 2.25th Term = $8 + (0.25)(15 - 8) = 9.75$

First Quartile value is 9.75

Second Quartile = $\{(n + 1)/2\}$ th term

$\Rightarrow$ Second Quartile = $(9 + 1)/2$ th term

$\Rightarrow$ Second Quartile = $\{10/2\}$ th term

$\Rightarrow$ Second Quartile = 5th term

5th term is 20

So the second Quartile value is 20.

Third Quartile = $3(n + 1)/4$ th term

$\Rightarrow$ Third Quartile = $(3(8 + 1)/4)$ th term

$\Rightarrow$ Third Quartile = $(27/4)$ th term

$\Rightarrow$ Third Quartile = 6.75th term

Thus, 6.75th = 6th term + $(0.75)(7\text{th} - 6\text{th})$

$\Rightarrow 6.75\text{th} = 25 + (0.75)(5) = 28.75$

So the third Quartile value is 28.75

Problem 5: What is the Interquartile Range for the data if the first quartile is 10 and the third quartile is 30cm?

Solution:

Given,

- $Q_1 = 10$

- $Q_3 = 30$

Interquartile range = $Q_3 - Q_1$

$\Rightarrow$ Interquartile range = $30 - 10$

Thus, Interquartile range is 20.

Problem 6: What is the Quartile Deviation for the data if the first quartile is 15 and the third quartile is 30cm?

Solution:

Given,

- $Q_1 = 15$

- $Q_3 = 30$

Quartile Deviation = $(Q_3 - Q_1)/2$

$\Rightarrow$ Quartile Deviation = $(30 - 15)/2$

$\Rightarrow$ Quartile Deviation = $15/2$

Thus, Quartile Deviation is 7.5

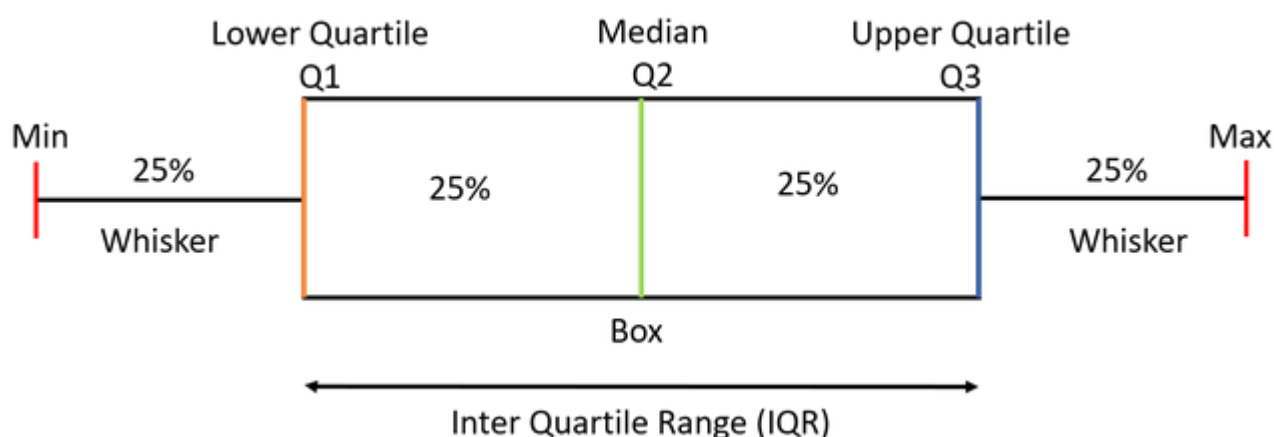
What is a Box Plot?

The idea of box plot was presented by John Tukey in 1970. He wrote about it in his book “Exploratory Data Analysis” in 1977. Box plot is also known as a whisker plot, box-and-whisker plot, or simply a box-and whisker diagram. Box plot is a graphical representation of the distribution of a dataset. It displays key summary statistics such as the [median](#), [quartiles](#), and potential [outliers](#) in a concise and visual manner. By using Box plot you can provide a summary of the distribution, identify potential and compare different datasets in a compact and visual manner.

Elements of Box Plot

A box plot gives a five-number summary of a set of data which is-

- **Minimum** – It is the minimum value in the dataset excluding the outliers.
- **First Quartile (Q1)** – 25% of the data lies below the First (lower) Quartile.
- **Median (Q2)** – It is the mid-point of the dataset. Half of the values lie below it and half above.
- **Third Quartile (Q3)** – 75% of the data lies below the Third (Upper) Quartile.
- **Maximum** – It is the maximum value in the dataset excluding the outliers.



The area inside the box (50% of the data) is known as the [Inter Quartile Range](#). The **IQR** is calculated as –

$$\text{IQR} = \text{Q3} - \text{Q1}$$

Outliers are the data points **below and above the lower and upper limit**. The lower and upper limit is calculated as –

$$\text{Lower Limit} = \text{Q1} - 1.5 \times \text{IQR}$$

$$\text{Upper Limit} = \text{Q3} + 1.5 \times \text{IQR}$$

The values below and above these limits are considered outliers and the minimum and maximum values are calculated from the points which lie under the lower and upper limit.

How to create a box plots?

Let us take a sample data to understand how to create a box plot.

Here are the runs scored by a cricket team in a league of 12 matches – **100, 120, 110, 150, 110, 140, 130, 170, 120, 220, 140, 110**.

To draw a box plot for the given data first we need to arrange the data in ascending order and then find the minimum, first quartile, median, third quartile and the maximum.

Ascending Order

100, 110, 110, 110, 120, 120, 130, 140, 140, 150, 170, 220

Median (Q2) = $(120 + 130) / 2 = 125$; Since there were even values

To find the First Quartile we take the first six values and find their median.

$$\mathbf{Q1 = (110+110)/2 = 110}$$

For the Third Quartile, we take the next six and find their median.

$$\mathbf{Q3 = (140+150)/2 = 145}$$

Note: If the total number of values is odd then we exclude the Median while calculating Q1 and Q3. Here since there were two central values we included them. Now, we need to calculate the Inter Quartile Range.

$$\mathbf{IQR = Q3-Q1 = 145-110 = 35}$$

We can now calculate the Upper and Lower Limits to find the minimum and maximum values and also the outliers if any.

$$\mathbf{Lower\ Limit = Q1-1.5*IQR = 110-1.5*35 = 57.5}$$

$$\mathbf{Upper\ Limit = Q3+1.5*IQR = 145+1.5*35 = 197.5}$$

So, the minimum and maximum between the range [57.5,197.5] for our given data are –

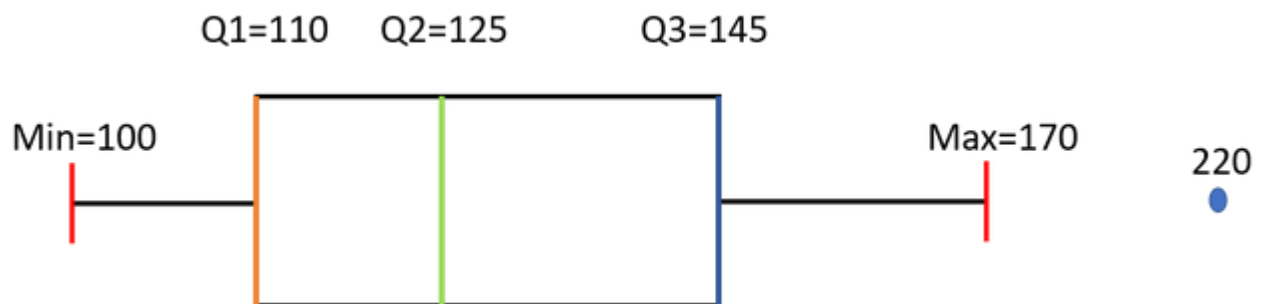
$$\mathbf{Minimum = 100}$$

$$\mathbf{Maximum = 170}$$

The outliers which are outside this range are –

$$\mathbf{Outliers = 220}$$

Now we have all the information, so we can draw the box plot which is as below-



We can see from the diagram that the Median is not exactly at the center of the box and one whisker is longer than the other. We also have one Outlier.

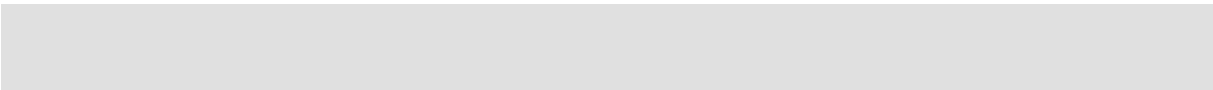
Use-Cases of Box Plot

a) If the Median is at the **center** of the Box and the **whiskers** are almost the **same on both the ends** then the data is **Normally Distributed**.

b) If the Median lies **closer to the First Quartile** and if the **whisker at the lower**

end is shorter (as in the above example) then it has a **Positive Skew (Right Skew)**.

c) If the Median lies **closer to the Third Quartile** and if the **whisker at the upper end is shorter** than it has a **Negative Skew (Left Skew)**.



Normal Distribution



Positive Skew



Negative Skew

